



Barker College

**2003
YEAR 12
TRIAL HSC
EXAMINATION**

MATHEMATICS

Staff Involved:

DATE : AM Wednesday 6 August

- PJR* • JML*
- RMH • GDH
- MRB • CFR
- BJR • GIC
- VAB • AJD
- LJP

160 copies

General Instructions

- **Reading time – 5 minutes**
- **Working time – 3 hours**
- **Write using blue or black pen**
- **Board-approved calculators may be used**
- **A Table of Standard Integrals is provided at the back of this paper**
- **ALL necessary working MUST be shown in every question**

Total marks - 120

- **Attempt Questions 1 - 10**
- **All questions are of equal value**
- **Answer EACH QUESTION on a NEW PIECE of lined paper**
- **Only write on ONE side of the lined paper**
- **Write your Barker Student Number at the top of each page of your answers**

Total marks - 120
Attempt Questions 1 - 10
All questions are of equal value

Answer each question on a SEPARATE piece of lined paper.

Question 1 (12 marks)	Use a NEW piece of lined paper.	Marks
(a) Simplify $\frac{6x + 15x^3}{3x}$		2
(b) Solve $4x < 3(x - 1)$		2
Graph your solution on a number line.		
(c) Find a primitive of $\sqrt{x^3} - 5$		2
(d) Express $\frac{\sqrt{6}}{\sqrt{6} - \sqrt{5}}$ in the form $a + b\sqrt{c}$, where a , b and c are integers.		2
(e) Using the table of standard integrals, find $\int \sec 4x \tan 4x \, dx$		1
(f) Find the exact value of $\log_e e^2$		1
(g) The price of an article for sale is \$160, which includes GST of 10%.		2
Calculate the price of the article without the GST.		

- (a) Solve the pair of simultaneous equations

2

$$y = 8 - 2x$$

$$x - y = 7$$

- (b) The diagram below shows the parallelogram $OABC$ with vertices $O(0, 0)$, $A(3, 5)$, $B(8, 6)$ and C .

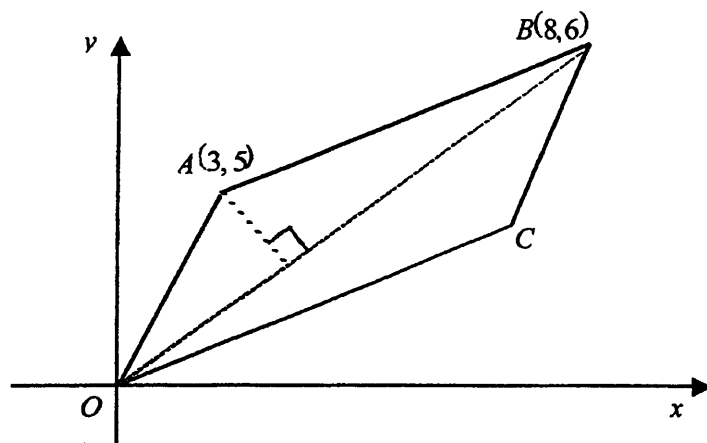


DIAGRAM
NOT TO SCALE

- (i) Write down the coordinates of the mid-point of OB .
- (ii) Find the coordinates of C .
- (iii) Show that the equation of the line OB is $3x - 4y = 0$.
- (iv) Show that the length of the interval OB is 10 units.
- (v) Calculate the perpendicular distance from A to the line OB .
- (vi) Calculate the area of the parallelogram $OABC$.

1

1

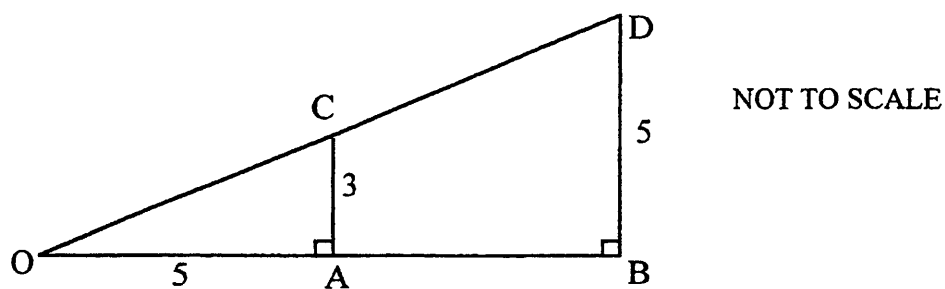
2

2

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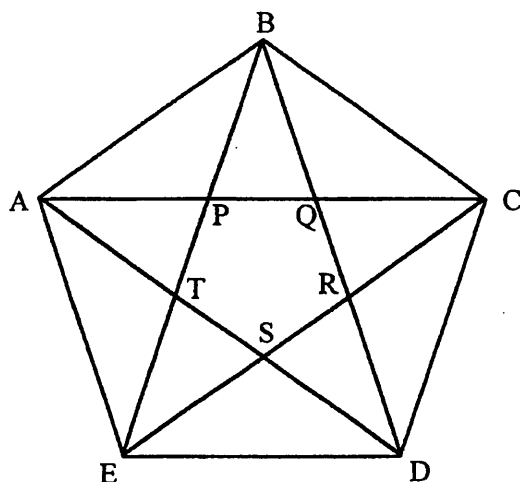
(a)



In the diagram above, AC is parallel to DB, $OA = 5\text{cm}$, $DB = 5\text{cm}$ and $AC = 3\text{cm}$.

- (i) Show that triangles OCA and ODB are similar. 3
- (ii) Hence, find the length of AB, giving reasons. 2

(b)



ABCDE is a regular pentagon with each side being equal in length.

Equal diagonals have been drawn between all vertices to form another smaller regular pentagon PQRST.

- (i) Find the size of $\angle CDE$. 1
- (ii) Show that triangles ADE and CDE are congruent. 3
- (iii) Find the size of $\angle DAE$, giving reasons. 1
- (iv) Hence, or otherwise, find the size of $\angle ATP$ giving reasons for your answer. 2

Question 4 (12 marks) Start a NEW piece of lined paper.

Marks

(a) Find the equation of the normal to the curve $y = x^3 - 3x + 1$ at the point (2, 3). 3

(b) Differentiate with respect to x :

(i) $\ln(3x - 4)^5$ 2

(ii) $3x^2 e^{x^3+1}$ 2

(c) Find:

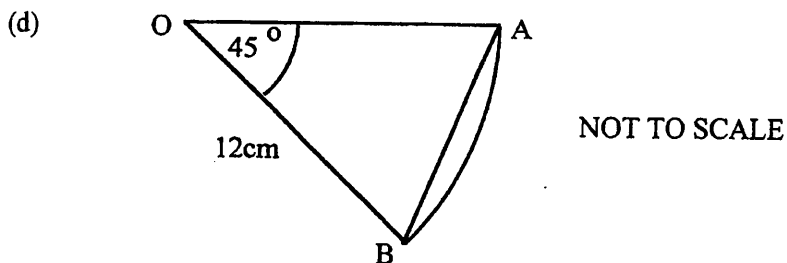
(i) $\int (x^2 + r^2) dx$ (where r is a constant) 2

(ii) $\int_3^4 \frac{2}{x-2} dx$ 3

Question 5 (12 marks) Start a NEW piece of lined paper.

Marks

- (a) Consider the parabola with equation $(x + 3)^2 = 6(y - 1)$
- (i) Find the coordinates of the vertex of the parabola 1
- (ii) Find the coordinates of the focus of the parabola 2
- (b) For the quadratic equation $x^2 - (k - 1)x + (k - 2) = 0$, find the values of k for which the roots are real and different. 3
- (c) Find y such that $\log_e y + \log_e 3 = 2.4$ 2
- (Give your answer as a decimal correct to two decimal places)



The diagram above shows a sector AOB of a circle with radius 12cm.

- (i) Find the length of the arc AB. 2
- (ii) Find the length of the straight line AB. 2

- (a) The first three partial sums of a series are $S_1 = 9$, $S_2 = 25$ and $S_3 = 50$ 2

Find the first three terms, T_1 , T_2 and T_3 , of this series.

- (b) The third term of a geometric series is 36 and the sixth term is 972

(i) Find the common ratio r 2

(ii) Find the first term a 1

- (c) A super-bouncy ball is dropped from a height and takes 1 second to hit the ground.
The ball then bounces, taking a further $1\frac{1}{2}$ seconds before it bounces on the ground again.
Each successive bounce takes $\frac{3}{4}$ of the previous amount of time to bounce again.
Find how long the ball will be in motion before coming to rest. 3

- (d) William starts playing the game Space Cadet Pinball and gets an initial high score of 858.
He then regularly plays this game and keeps a record of the improvement in his high scores at the end of each week. These are recorded in the table below.

Week	1	2	3	4	5	6	...	14	15	...
Improvement	298	281	264	247	230	213	...	77	60	...

- (i) How many weeks will it take for William to reach his overall maximum score? 2
- (ii) What will be William's overall maximum score? 2

- (a) A resistance force, F , is related to the speed, v , by the function

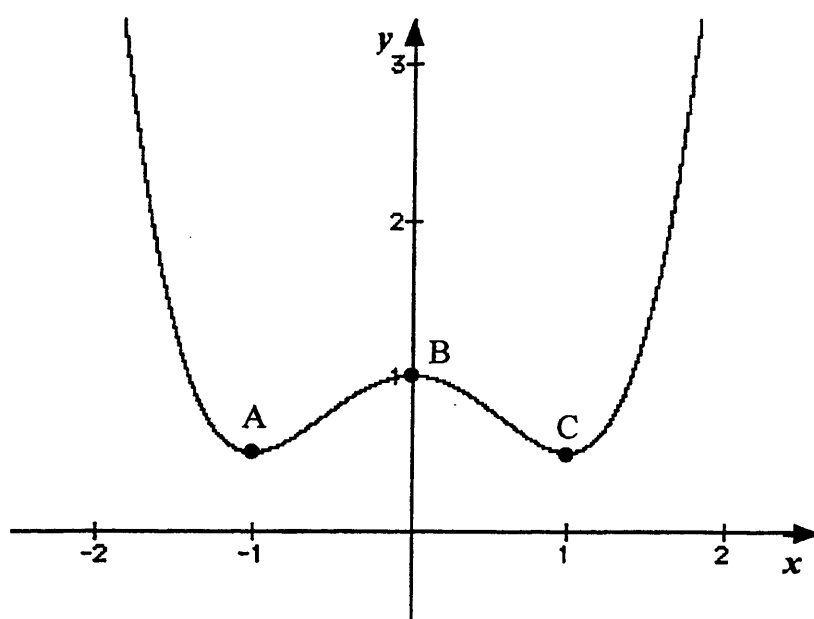
$$F(v) = v^2 + \frac{16}{v}$$

- (i) Find $F'(v)$ 2
- (ii) This resistance force is minimised when $F'(v) = 0$ 2

Find the speed at which this resistance force is minimised.

- (b) The graph of $y = \frac{1}{2}x^4 - x^2 + 1$ is sketched below.

The points A, B and C are the stationary points of this curve.



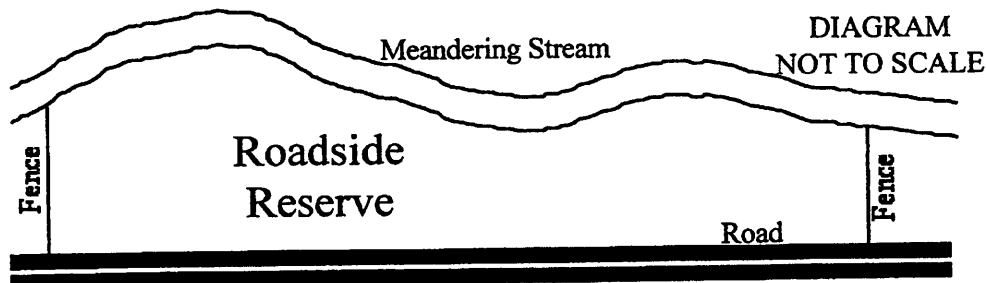
- (i) Find the coordinates of the points A, B and C. 3
- (ii) For what values of x is this curve concave down? 3
- Give reasons for your answer.

- (iii) Using Part (i), draw a rough sketch of the gradient function, $\frac{dy}{dx}$, of this curve. 2

Identify any points on this sketch where the concavity of the curve changes.

- (a) Find the equation of the curve that passes through the point (5, 4) and has a gradient function of $2x - 6$ 2

- (b) A roadside reserve is bounded by a straight road, a meandering stream and two straight fences to neighbouring farms. The border along the road is 120 metres.



The width of the roadside reserve from the road-edge to the stream-bank, measured at 20 metre intervals along the road is given in the table below.

Distance along road (m)	0	20	40	60	80	100	120
Width of roadside reserve (m)	20	30	22	16	18	18	14

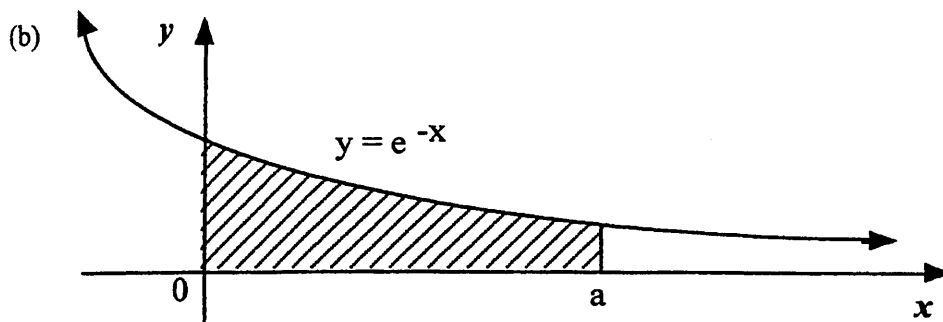
Use Simpson's Rule to approximate the area of this roadside reserve. 3

- (c) (i) On the same axes sketch $y = x^2 - x$ and $y = x + 3$ 2
- (ii) Find the x values of the points of intersection of these two curves. 2
- (iii) Hence, find the area between these two curves. 3

(a) (i) Without using calculus, sketch the curve $y = e^x - 2$ 2

(ii) On the same sketch, find, graphically, the number of solutions of the equation 2

$$e^x - x - 2 = 0$$



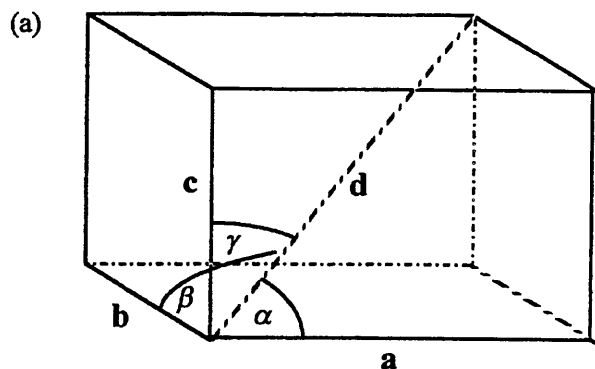
(i) What is the volume of the solid formed when the shaded area is rotated completely around the x -axis? 2

(ii) What is the limit of this volume as $a \rightarrow \infty$? 1

(c) Amelia has borrowed \$5000 at the beginning of 2003.
 The debt is to be repaid by equal annual installments of \$1200.
 The first installment is to be repaid at the beginning of 2004.
 Interest at the rate of 18% p.a. is calculated at the beginning of each year on the balance owing at the end of the previous year.
 This interest is then added to the balance of the debt before a repayment is made.

(i) Calculate the amount owing on the debt after the first repayment is made at the beginning of 2004. 2

(ii) Amelia wants to clear her debt at the beginning of 2010.
 How much extra will she have to repay at the beginning of 2010 (after making her normal repayment) in order for her to do this? 3



A rectangular box has edges of length a , b and c .
A diagonal of length d is drawn through the box between opposite corners as shown.

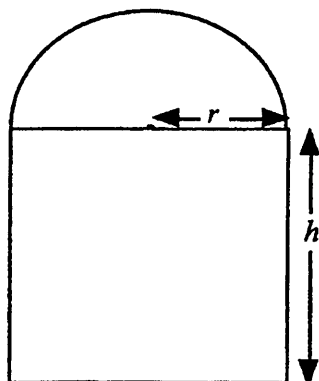
The three different angles between this diagonal and the three edges a , b and c of the box are labeled α , β and γ respectively.

- (i) Express d in terms of a , b and c . 2

- (ii) Hence, or otherwise, show that the angles α , β and γ obey the identity 3

$$\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma = 1$$

- (b) A window in the chapel has been damaged by a storm and needs to be replaced.



It is in the shape of a rectangle surmounted by a semi-circle, as shown.

Let the radius of the semi-circle be r metres and the height of the rectangle be h metres.

- (i) Given that the perimeter of the window is to be 10π metres, show that 2

$$h = 5\pi - r - \frac{\pi r}{2}$$

- (ii) Hence, show that the area of the window is given by the formula 1

$$A = 10\pi r - 2r^2 - \frac{1}{2}\pi r^2$$

- (ii) Hence, find the radius of the circle for which the area of the window is to be a maximum. 4

END OF PAPER

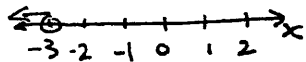
2003 2 Unit Mathematics Trial HSC Solutions

Question 1

(a) $\frac{3x(2+5x^2)}{3x}$

$= 2 + 5x^2$

(b) $4x < 3x - 3$
 $x < -3$



(c) $\int x^{\frac{5}{2}} - 5 dx$
 $= \frac{2x^{\frac{7}{2}}}{\frac{7}{2}} - 5x + C$

(d) $\frac{\sqrt{6}}{(\sqrt{6}-\sqrt{5})} \times \frac{(\sqrt{6}+\sqrt{5})}{(\sqrt{6}+\sqrt{5})}$
 $= \frac{6 + \sqrt{30}}{6 - 5}$
 $= 6 + \sqrt{30}$

(e) $\frac{1}{4} \sec 4x + C$

(f) $2 \log_e e$
 $= 2$

(g) $110\% = \$160$
 $10\% = \frac{160}{11}$
 $\therefore 100\% = \frac{160}{11} \times 10$
 $= \$145.45$

Question 2

(a) $\begin{cases} 2x + y = 8 \\ x - y = 7 \end{cases} \therefore 3x = 15$
 $x = 5$
 $\therefore 10 + y = 8 \quad y = -2$
 $x = 5, y = -2$

(b)(i) Midpt of AB = $(\frac{8}{2}, \frac{6}{2})$
 $= (4, 3)$

(ii) $C \Rightarrow (8-3, 6-5)$
 C is $(5, 1)$

(iii) $m = \frac{6-0}{8-0} = \frac{3}{4}$
Eqn is $y - 0 = \frac{3}{4}(x - 0)$

$\therefore 4y = 3x$

$\therefore 3x - 4y = 0$

(iv) $d_{OB} = \sqrt{(8-0)^2 + (6-0)^2}$
 $= \sqrt{64 + 36}$
 $= \sqrt{100} = 10 \text{ units}$

(v) $(3, 5) \quad 3x - 4y = 0$

$d = \frac{|3 \times 3 + 5 \times -4 + 0|}{\sqrt{3^2 + (-4)^2}}$
 $= \frac{|9 - 20|}{\sqrt{9 + 16}}$
 $= \frac{|-11|}{\sqrt{25}} = \frac{11}{5} \text{ units}$

(vi) Area of $\triangle OAB$
 $= 2 \times \text{Area of } \triangle OAB$
 $= 2 \times (\frac{1}{2} \times 10 \times \frac{11}{5})$
 $= 22 \text{ units}^2$

Question 3

(a) $\hat{OAC} = \hat{OBD} = 90^\circ$
(right angles given)

$\hat{COA} = \hat{DOB}$ (common angle)

$\hat{OCA} = \hat{ODB}$ (corres. $\angle$ s on $\parallel$ lines are equal)

$\therefore \triangle OCA \parallel \triangle ODB$ (equal angles)

(ii) $\frac{AC}{BD} = \frac{OA}{OB}$ (sides of similar $\triangle$ s in same ratio)

$\therefore \frac{3}{5} = \frac{5}{5 + AB}$

$\therefore 15 + 3AB = 25$

$\therefore 3AB = 10$

$\therefore AB = \frac{10}{3} \text{ cm}$

(b)(i) Angle sum of $\triangle CDE$ $= (5-2) \times 180$
 $= 3 \times 180$
 $= 540^\circ$

$\therefore \hat{CDE} = \frac{540}{5}$
 $= 108^\circ$

(ii) ED is common
 $AE = CD$ (equal sides of regular pentagon)

$\hat{CDE} = \hat{AED}$ (equal angles of regular pentagon)

$\therefore \triangle ADE \equiv \triangle CDE$ (SAS)

(iii) $\triangle ADE$ is isos. $\triangle$ ($AE = DE$ equal sides of pentagon)
thus $\hat{DAE} = \hat{ADE}$
 $= \frac{180 - 108}{2}$ (Equal base $\angle$ s of isos $\triangle$)
 $= 36^\circ$

(iv) $\hat{P\hat{T}S} = 108^\circ$ (equal angles of regular pentagon)
 $\therefore \hat{ATP} = 180 - 108$ (straight line angle)
 $= 72^\circ$

Question 4

(a) $y = x^3 - 3x + 1$

$\frac{dy}{dx} = 3x^2 - 3$

When $x = 2, y' = 3 \times 2^2 - 3$

$\therefore m = 9$

$\therefore$ Grad of normal $= -\frac{1}{9}$

Eqn of normal is

$y - 3 = -\frac{1}{9}(x - 2)$

$\therefore 9y - 27 = -x + 2$

$\therefore x + 9y - 29 = 0$

(b)(i) $y = \ln(3x - 4)$
 $y = 5 \ln(3x - 4)$

$\therefore y' = 5 \times \frac{3}{3x - 4}$
 $= \frac{15}{3x - 4}$

(ii) $y = 3x^2 e^{x^3 + 1}$

$u = 3x^2 \quad u' = 6x$

$v = e^{x^3 + 1} \quad v' = 3x^2 e^{x^3 + 1}$

$\frac{dy}{dx} = 3x^2 \times 3x^2 e^{x^3 + 1} + 6x \times e^{x^3 + 1}$

$= 9x^4 e^{x^3 + 1} + 6x e^{x^3 + 1}$

(c)(i) $\int x^3 + r^2 dx$

$= \frac{x^4}{4} + r^2 x + C$

(ii) $2 \int_3^4 \frac{1}{x-2} dx$

$= 2 [\ln(x-2)]_3^4$

$= 2 [\ln 2 - \ln 1]$

$= 2 \ln 2$

Question 5

(a)(i) Vertex $\Rightarrow (-3, 1)$



$4a = 6$
 $\therefore a = \frac{3}{2}$

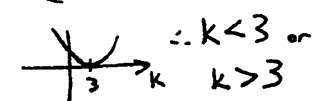
$\therefore$ Focus $\Rightarrow (-3, 2\frac{1}{2})$

(b) $\Delta > 0$ if roots real & different

$\Delta = (k-1)^2 - 4 \times 1 \times (k-2)$
 $= k^2 - 2k + 1 - 4k + 8$
 $= k^2 - 6k + 9$

$\therefore k^2 - 6k + 9 > 0$

$\therefore (k-3)^2 > 0$



(c) $\log_e 3y = 2.4$

$\therefore 3y = e^{2.4}$

$\therefore y = \frac{e^{2.4}}{3}$
 ≈ 3.67

(d)(i) $45^\circ = \frac{\pi}{4}$ radians

$l = r\theta \Rightarrow l = 12 \times \frac{\pi}{4}$
 $\therefore l = 3\pi \text{ cm}$
 $\approx 9.42 \text{ cm}$

(ii) $OA = OB = 12$

$c^2 = 12^2 + 12^2 - 2 \times 12 \times 12 \cos 120^\circ$

$c^2 = 84 - 3532 \dots$

$\therefore c \approx 9.18 \text{ cm}$

Question 6

(a) $S_1 = T_1 = 9$

$S_2 = T_1 + T_2$

$\therefore 25 = 9 + T_2$

$\therefore T_2 = 16$

$S_3 = T_1 + T_2 + T_3$

$\therefore 50 = 9 + 16 + T_3$

$\therefore T_3 = 25$

(b) $T_3 = 36, T_6 = 972$

(i) $ar^5 = 972$
 $ar^2 = 36$

$\frac{ar^5}{ar^2} = \frac{972}{36}$

$\therefore r^3 = 27$

$\therefore r = 3$

(ii) $a \times 3^2 = 36$

$\therefore a = \frac{36}{9}$

$\therefore a = 4$

(c) $1 + \frac{3}{2} + \frac{3}{2} \times \frac{3}{4} + \frac{3}{2} \times \frac{3}{4} \times \frac{3}{4}$

= 1 + sum of infinite G.P.

with $a = \frac{3}{2}, r = \frac{3}{4}$

= $1 + \frac{\frac{3}{2}}{1 - \frac{3}{4}}$

= $1 + 6$

= 7 seconds comes to rest

(d)(i) $a = 298, d = -17$

Overall max score when $T_n = 0$

$0 = 298 + (n-1)(-17)$

$0 = 298 - 17n + 17$

$17n = 315$

$n = 18.5$ i.e. in 19 weeks

(ii) $T_{18} = 298 + 17(-17)$
 $= 9$

$\therefore$ Max score = $858 + S_{18} + 9$

$S_{18} = \frac{18}{2} [2 \times 298 + 17(-17)]$
 $= 9(596 - 289)$

$S_{18} = 2763$

$\therefore$ Max Score = 3630

Question 7

(a)(i) $F(v) = v^2 + 16v^{-1}$

$F'(v) = 2v - 16v^{-2}$

(ii) $0 = 2v - \frac{16}{v^2}$

$\therefore \frac{16}{v^2} = 2v$

$\therefore 16 = 2v^3$

$\therefore v^3 = 8$

$\therefore v = 2$ when force is minimised

(b)(i) $y = \frac{1}{2}x^4 - x^2 + 1$

$\frac{dy}{dx} = 2x^3 - 2x$

stat pt $\Rightarrow \frac{dy}{dx} = 0$

$\therefore 2x^3 - 2x = 0$

$\therefore 2x(x^2 - 1) = 0$

$\therefore 2x(x+1)(x-1) = 0$

$\therefore x = 0, -1, 1$

when $x = 0, y = 1$ B is (0, 1)

when $x = \pm 1, y = \frac{1}{2} - 1 + 1$
 $= \frac{1}{2}$

$\therefore A = (-1, \frac{1}{2})$ $C = (1, \frac{1}{2})$

(ii) $y'' = 6x^2 - 2$

Concave down $\Rightarrow y'' < 0$

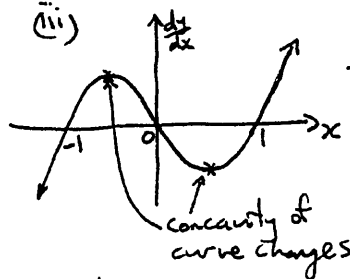
$\therefore 6x^2 - 2 < 0$

$x^2 < \frac{1}{3} < 0$

$(x - \frac{1}{\sqrt{3}})(x + \frac{1}{\sqrt{3}}) < 0$

$\therefore -\frac{1}{\sqrt{3}} < x < \frac{1}{\sqrt{3}}$

(iii)



Question 8

(a) $\frac{dy}{dx} = 2x - 6$

$y = x^2 - 6x + C$

Subst. (5, 4) $\Rightarrow 4 = 25 - 30 + C$

$\therefore 4 = -5 + C$

$C = 9$

curve is

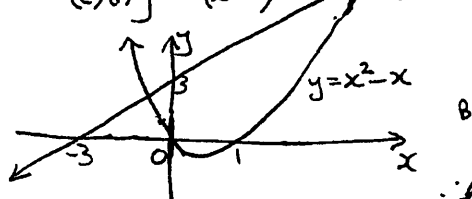
$y = x^2 - 6x + 9$

(b) $h = 20$

Area $\div \frac{20}{3} [20 + 14] + 4 \times (30 + 16 + 18)$
 $+ 2 \times (22 + 18)]$

$\therefore$ Area $\div 2466 \frac{2}{3} \text{ m}^2$

(c)(i) $y = x(x-1)$



(iii) $y = x^2 - x$
 $y = x+3$

$\therefore x^2 - x = x+3$

$\therefore x^2 - 2x - 3 = 0$

$\therefore (x-3)(x+1) = 0$

$\therefore x = 3$ or -1

(ii) $A = \int_{-1}^3 (x+3) - (x^2-x) dx$

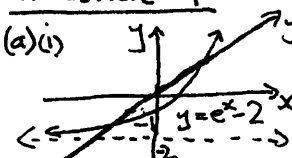
$= \int_{-1}^3 3 + 2x - x^2 dx$

$= [3x + x^2 - \frac{x^3}{3}]_{-1}^3$

$= (9 + 9 - \frac{27}{3}) - (-3 + 1 - \frac{1}{3})$

$= 10 \frac{2}{3} \text{ units}^2$

Question 9



(i) $e^x - 2 = x$

$y = e^x - 2$ and $y = x$

$\therefore$ 2 solutions to eqn

(b)(i) $V = \pi \int_0^a (e^{-x})^2 dx$

$= \pi \int_0^a e^{-2x} dx$

$= \pi [\frac{e^{-2x}}{-2}]_0^a$

$= -\frac{\pi}{2} (e^{-2a} - e^0)$

$= -\frac{\pi}{2} (e^{-2a} - 1) \text{ units}^3$

(ii) $V = -\frac{\pi}{2} (\frac{1}{e^{2a}} - 1)$

As $a \rightarrow \infty, \frac{1}{e^{2a}} \rightarrow 0$

$\therefore V \rightarrow \frac{\pi}{2} \text{ units}^3$

(c)(i) $A_1 = 5000 \times 1.18 - 1200$
 $= \$4700$

(ii) $A_2 = A_1 \times 1.18 - 1200$

$= 5000 \times 1.18^2 - 1200(1.18 + 1)$

$A_3 = A_2 \times 1.18 - 1200$

$= 5000 \times 1.18^3 - 1200(1.18^2 + 1.18 + 1)$

Begin of 2010 $\Rightarrow A_7 = 5000 \times 1.18^7 - 1200(1.18^6 + \dots + 1.18 + 1)$

$A_7 = 5000 \times 1.18^7 - 1200 \times \frac{1 \times (1.18^7 - 1)}{1.18 - 1}$
 $= \$1357.54$

Question 10

(a)(i) Diagonal of = $\sqrt{a^2 + b^2}$
 base

$\therefore d^2 = c^2 + (\sqrt{a^2 + b^2})^2$

$\therefore d^2 = a^2 + b^2 + c^2$

$\therefore d = \sqrt{a^2 + b^2 + c^2}$

(ii) $\cos \alpha = \frac{a}{d}, \cos \beta = \frac{b}{d}$ and $\cos \gamma = \frac{c}{d}$

LHS = $\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma$

$= \frac{a^2}{d^2} + \frac{b^2}{d^2} + \frac{c^2}{d^2}$

$= \frac{a^2 + b^2 + c^2}{d^2}$

$= \frac{d^2}{d^2} = 1 = \text{RHS}$

(b)(i) $10\pi = 2h + 2r + \pi r$

$\therefore 2h = 10\pi - 2r - \pi r$

$\therefore h = 5\pi - r - \frac{\pi r}{2}$

(ii) $A = 2rh + \frac{1}{2}\pi r^2$

$\therefore A = 2r(5\pi - r - \frac{\pi r}{2}) + \frac{\pi r^2}{2}$

$\therefore A = 10\pi r - 2r^2 - \pi r^2 + \frac{\pi r^2}{2}$

$\therefore A = 10\pi r - 2r^2 - \frac{\pi r^2}{2}$

(iii) $A' = 10\pi - 4r - \pi r$

Max when $A' = 0 \Rightarrow$

$\therefore 0 = 10\pi - 4r - \pi r$

$\therefore \pi r + 4r = 10\pi$

$\therefore r(\pi + 4) = 10\pi$

$\therefore r = \frac{10\pi}{\pi + 4} (\approx 4.4)$

r	4	4.4	5
A'	2.8	0	-4.3

Thus, maximum when $r = \frac{10\pi}{\pi + 4}$ metres